

# 1Q) DATASET 1

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EXPERIMENT 1A.py - C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 1A.py (3.14.3)
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# FIND-S Algorithm in Python

# Training Data (Play Tennis Dataset)
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0]) - 1)

print("Training Data:")
for row in data:
    print(row)

print("\nInitial Hypothesis:")
print(hypothesis)

# Apply FIND-S Algorithm
for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'

        print("\nUpdated Hypothesis:")
        print(hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> == RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 1A.py ==
Training Data:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes']
['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes']
['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No']
['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Updated Hypothesis:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Updated Hypothesis:
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Updated Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']

Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
>>>
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# DATASET 2

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EXPERIMENT 1 B.py - C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 B.py (3.14.3)
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# FIND-S Algorithm - Loan Approval Dataset

# Training Data
data = [
    ['High', 'Good', 'Permanent', 'Yes', 'Yes', 'Approved'],
    ['High', 'Good', 'Permanent', 'No', 'Middle', 'No'],
    ['Low', 'Good', 'Temporary', 'No', 'Young', 'No'],
    ['Medium', 'Good', 'Permanent', 'Yes', 'Middle', 'Approved'],
    ['High', 'Average', 'Temporary', 'Yes', 'Old', 'No'],
    ['High', 'Good', 'Permanent', 'Yes', 'Middle', 'Approved']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0]) - 1)

print("Training Data:\n")
for row in data:
    print(row)

print("\nInitial Hypothesis:")
print(hypothesis)

# FIND-S Algorithm
for row in data:
    if row[-1] == "Approved":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'

        print("\nUpdated Hypothesis:")
        print(hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 B.py =====
Training Data:
['High', 'Good', 'Permanent', 'Yes', 'Yes', 'Approved']
['High', 'Good', 'Permanent', 'No', 'Middle', 'No']
['Low', 'Good', 'Temporary', 'No', 'Young', 'No']
['Medium', 'Good', 'Permanent', 'Yes', 'Middle', 'Approved']
['High', 'Average', 'Temporary', 'Yes', 'Old', 'No']
['High', 'Good', 'Permanent', 'Yes', 'Middle', 'Approved']

Initial Hypothesis:
['0', '0', '0', '0', '0']

Updated Hypothesis:
['High', 'Good', 'Permanent', 'Yes', 'Yes']

Updated Hypothesis:
['?', 'Good', 'Permanent', 'Yes', '?']

Updated Hypothesis:
['?', 'Good', 'Permanent', 'Yes', '?']

Final Most Specific Hypothesis:
['?', 'Good', 'Permanent', 'Yes', '?']
>>>
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# DATASET 3

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EXPERIMENT 1 C.py - C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 C.py (3.14.3)
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# FIND-S Algorithm - Student Placement Dataset

# Training Data
data = [
    ['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],
    ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],
    ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],
    ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],
    ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],
    ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0]) - 1)

print("Training Data:\n")
for row in data:
    print(row)

print("\nInitial Hypothesis:")
print(hypothesis)

# FIND-S Algorithm
for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'
                break
        print("\nUpdated Hypothesis:")
        print(hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 C.py =====
Training Data:

[['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],
 ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],
 ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],
 ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],
 ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],
 ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes']]

Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Updated Hypothesis:
['High', 'Good', 'Yes', 'Good', 'High']

Updated Hypothesis:
['High', '?', 'Yes', 'Good', 'High']

Updated Hypothesis:
['High', '?', 'Yes', '?', 'High']

Updated Hypothesis:
['High', '?', 'Yes', '?', '?']

Final Most Specific Hypothesis:
['High', '?', 'Yes', '?', '?']

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# DATASET 4

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EXPERIMENT 1 D.py - C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 D.py (3.14.3)
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# FIND-S Algorithm - Disease Diagnosis Dataset

# Training Data
data = [
    ['Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['Yes', 'Yes', 'No', 'Yes', 'Positive'],
    ['No', 'Yes', 'Yes', 'No', 'Negative'],
    ['Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['No', 'No', 'Yes', 'No', 'Negative'],
    ['Yes', 'Yes', 'No', 'Yes', 'Positive']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0]) - 1)

print("Training Data:\n")
for row in data:
    print(row)

print("\nInitial Hypothesis:")
print(hypothesis)

# FIND-S Algorithm
for row in data:
    if row[-1] == "Positive":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'
                break
        print("\nUpdated Hypothesis:")
        print(hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 D.py =====
Training Data:

[['Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
 ['Yes', 'Yes', 'No', 'Yes', 'Positive'],
 ['No', 'Yes', 'Yes', 'No', 'Negative'],
 ['Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
 ['No', 'No', 'Yes', 'No', 'Negative'],
 ['Yes', 'Yes', 'No', 'Yes', 'Positive']]

Initial Hypothesis:
['0', '0', '0', '0']

Updated Hypothesis:
['Yes', 'Yes', 'Yes', 'Yes']

Updated Hypothesis:
['Yes', 'Yes', '?', 'Yes']

Updated Hypothesis:
['Yes', 'Yes', '?', 'Yes']

Updated Hypothesis:
['Yes', 'Yes', '?', 'Yes']

Final Most Specific Hypothesis:
['Yes', 'Yes', '?', 'Yes']

>>>
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## DATASET 5

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EXPERIMENT 1 E.py - C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 E.py (3.14.3)
File Edit Format Run Options Window Help

# FIND-S Algorithm - Email Spam Detection Dataset

# Training Data
data = [
    ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes'],
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes'],
    ['No', 'No', 'No', 'No', 'No', 'No'],
    ['Yes', 'No', 'Yes', 'No', 'Yes', 'No'],
    ['No', 'Yes', 'No', 'Yes', 'No', 'No'],
    ['Yes', 'Yes', 'Yes', 'No', 'No', 'Yes']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0]) - 1)

print("Training Data:\n")
for row in data:
    print(row)

print("\nInitial Hypothesis:")
print(hypothesis)

# FIND-S Algorithm
for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'

        print("\nUpdated Hypothesis:")
        print(hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/mohit/OneDrive/Documents/EXPERIMENT 1 E.py =====
Training Data:

['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes']
['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes']
['No', 'No', 'No', 'No', 'No', 'No']
['Yes', 'No', 'Yes', 'No', 'Yes', 'No']
['No', 'Yes', 'No', 'Yes', 'No', 'No']
['Yes', 'Yes', 'Yes', 'No', 'No', 'Yes']

Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Updated Hypothesis:
['Yes', 'Yes', 'Yes', 'No', 'Yes']

Updated Hypothesis:
['Yes', 'Yes', 'Yes', '?', 'Yes']

Updated Hypothesis:
['Yes', 'Yes', 'Yes', '?', '?']

Final Most Specific Hypothesis:
['Yes', 'Yes', 'Yes', '?', '?']
>>>
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## 2Q) DATASET 1

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candidate_elimination.py.py - C:/Users/mohit/OneDrive/Documents/New folder (2)/candidate_elimination.py.py (3.14.3)
File Edit Format Run Options Window Help
for i, h in enumerate(concepts):
    if target[i].lower() == "yes":
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                specific_h[x] = '?'
                general_h[x][x] = '?'
    else:
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'

    print("\nStep", i + 1)
    print("Specific Hypothesis:")
    print(specific_h)
    print("General Hypothesis:")
    print(g)

# Remove duplicate rows
gh = []
for i in general_h:
    if i != ['?'] for j in range(len(specific_h))]:
        gh.append(i)

return specific_h, gh

s_final, g_final = candidate_elimination(concepts, target)

print("\n-----")
print("Final Specific Hypothesis")
print(s_final)

print("\nFinal General Hypothesis")
for g in g_final:
    print(g)

IDLE Shell 3.14.3
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Step 6
Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
General Hypothesis:
[['Sunny', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', 'Strong', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']]

Step 7
Specific Hypothesis:
['?', 'Warm', '?', 'Strong', '?', '?']
General Hypothesis:
[['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', 'Strong', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']]

Step 8
Specific Hypothesis:
['?', 'Warm', '?', 'Strong', '?', '?']
General Hypothesis:
[['?', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']]

-----
Final Specific Hypothesis
['?', 'Warm', '?', 'Strong', '?', '?']

Final General Hypothesis
['?', 'Warm', '?', '?', '?', '?']
>>>
```

# DATASET 2

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EXPERIMENT 2 B.py - C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 2 B.py (3.14.3)
File Edit Format Run Options Window Help
# Candidate Elimination Algorithm - Loan Approval Dataset

data = [
    ['High', 'Good', 'Permanent', 'Yes', 'Young', 'Yes'],
    ['High', 'Good', 'Permanent', 'No', 'Middle', 'No'],
    ['Low', 'Poor', 'Temporary', 'No', 'Young', 'No'],
    ['Medium', 'Good', 'Permanent', 'Yes', 'Middle', 'Yes'],
    ['Medium', 'Good', 'Permanent', 'No', 'Young', 'Yes'],
    ['Low', 'Good', 'Permanent', 'Yes', 'Middle', 'Yes'],
    ['Low', 'Poor', 'Temporary', 'Yes', 'Old', 'No']
]

concepts = [row[:-1] for row in data]
target = [row[-1] for row in data]

def candidate_elimination(concepts, target):
    specific_h = concepts[0].copy()
    general_h = [['?'] for i in range(len(specific_h))]
    for i in range(len(specific_h)):
        print("Initial Specific Hypothesis:")
        print(specific_h)
        print("\nInitial General Hypothesis:")
        for g in general_h:
            print(g)

    for i, h in enumerate(concepts):
        if target[i] == "Yes":
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = '?'
                    general_h[x][x] = '?'
        else:
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
                else:
                    general_h[x][x] = '?'

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 2 B.py ==
Initial Specific Hypothesis:
['High', 'Good', 'Permanent', 'Yes', 'Young']

Initial General Hypothesis:
[['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']]

Step 1
Specific Hypothesis: ['High', 'Good', 'Permanent', 'Yes', 'Young']
General Hypothesis:
[['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']]

Step 2
Specific Hypothesis: ['High', 'Good', 'Permanent', 'Yes', 'Young']
General Hypothesis:
[['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', 'Yes', '?']
['?', '?', '?', '?', 'Young']]

Step 3
Specific Hypothesis: ['High', 'Good', 'Permanent', 'Yes', 'Young']
General Hypothesis:
[['High', '?', '?', '?', '?']
['?', 'Good', '?', '?', '?']
['?', '?', 'Permanent', '?', '?']
['?', '?', '?', 'Yes', '?']
['?', '?', '?', '?', 'Young']]
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print(specific_h)

print("\nInitial General Hypothesis:")
for g in general_h:
    print(g)

for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                specific_h[x] = '?'
                general_h[x][x] = '?'
    else:
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'

    print("\nStep", i + 1)
    print("Specific Hypothesis:", specific_h)

    print("General Hypothesis:")
    for g in general_h:
        print(g)

gh = []
for g in general_h:
    if g != ['?'] for i in range(len(specific_h))]:
        gh.append(g)

return specific_h, gh

i_final, g_final = candidate_elimination(concepts, target)

print("\nFinal Specific Hypothesis:")
print(s_final)

print("\nFinal General Hypothesis:")
for g in g_final:
    print(g)

```

```

[ '?', '?', '?', '?', '?' ]
[ '?', 'Good', '?', '?', '?' ]
[ '?', '?', 'Permanent', '?', '?' ]
[ '?', '?', '?', 'Yes', '?' ]
[ '?', '?', '?', '?', '?' ]

Step 5
Specific Hypothesis: ['?', 'Good', 'Permanent', '?', '?']
General Hypothesis:
[ '?', '?', '?', '?', '?' ]
[ '?', 'Good', '?', '?', '?' ]
[ '?', '?', 'Permanent', '?', '?' ]
[ '?', '?', '?', '?', '?' ]

Step 6
Specific Hypothesis: ['?', 'Good', 'Permanent', '?', '?']
General Hypothesis:
[ '?', '?', '?', '?', '?' ]
[ '?', 'Good', '?', '?', '?' ]
[ '?', '?', 'Permanent', '?', '?' ]
[ '?', '?', '?', '?', '?' ]

Step 7
Specific Hypothesis: ['?', 'Good', 'Permanent', '?', '?']
General Hypothesis:
[ '?', '?', '?', '?', '?' ]
[ '?', 'Good', '?', '?', '?' ]
[ '?', '?', 'Permanent', '?', '?' ]
[ '?', '?', '?', '?', '?' ]

Final Specific Hypothesis:
[ '?', 'Good', 'Permanent', '?', '?' ]

Final General Hypothesis:
[ '?', 'Good', '?', '?', '?' ]
[ '?', '?', 'Permanent', '?', '?' ]

```

# DATASET 3

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EXPERIMENT 2 C.py - C:\Users\mohit\OneDrive\Documents\New folder (2)\EXPERIMENT 2 C.py (3.14.3)
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# Candidate Elimination Algorithm - Student Placement Dataset

data = [
    ['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],
    ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],
    ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],
    ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],
    ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],
    ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes'],
    ['Medium', 'Good', 'Yes', 'Good', 'High', 'Yes'],
    ['Low', 'Average', 'No', 'Poor', 'Medium', 'No']
]

concepts = [row[:-1] for row in data]
target = [row[-1] for row in data]

def candidate_elimination(concepts, target):
    specific_h = concepts[0].copy()
    general_h = [['?' for i in range(len(specific_h))]]

    print("Initial Specific Hypothesis:")
    print(specific_h)

    print("\nInitial General Hypothesis:")
    for g in general_h:
        print(g)

    for i, h in enumerate(concepts):
        if target[i] == "Yes":
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = '?'
                    general_h[x][x] = '?'
            else:
                for x in range(len(specific_h)):
                    if h[x] != specific_h[x]:
                        general_h[x][x] = specific_h[x]
                    else:
                        general_h[x][x] = '?'

    return specific_h, general_h

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```
print("\nInitial General Hypothesis:")
for g in general_h:
    print(g)

for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                specific_h[x] = '?'
                general_h[x][x] = '?'
    else:
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'

    print("\nStep", i + 1)
    print("Specific Hypothesis:", specific_h)
    print("General Hypothesis:")
    for g in general_h:
        print(g)

gh = []
for g in general_h:
    if g != ['?'] for i in range(len(specific_h)):
        gh.append(g)

return specific_h, gh

s_final, g_final = candidate_elimination(concepts, target)

print("\nFinal Specific Hypothesis:")
print(s_final)

print("\nFinal General Hypothesis:")
for g in g_final:
    print(g)
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General Hypothesis:
['High', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', 'High']

Step 6
Specific Hypothesis: ['High', '?', 'Yes', '?', '?']
General Hypothesis:
['High', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']

Step 7
Specific Hypothesis: ['?', '?', 'Yes', '?', '?']
General Hypothesis:
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']

Step 8
Specific Hypothesis: ['?', '?', 'Yes', '?', '?']
General Hypothesis:
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']

Final Specific Hypothesis:
['?', '?', 'Yes', '?', '?']

Final General Hypothesis:
['?', '?', 'Yes', '?', '?']

>>>
```

# DATASET 4

```
EXPERIMENT 2 D.py - C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 2 D.py (3.14.3)
File Edit Format Run Options Window Help

# Candidate Elimination Algorithm - Disease Diagnosis Dataset

data = [
    ['Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['Yes', 'Yes', 'No', 'Yes', 'Positive'],
    ['No', 'Yes', 'Yes', 'No', 'Negative'],
    ['Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['No', 'No', 'Yes', 'No', 'Negative'],
    ['Yes', 'Yes', 'No', 'Yes', 'Positive'],
    ['Yes', 'No', 'Yes', 'Yes', 'Positive'],
    ['No', 'Yes', 'No', 'No', 'Negative']
]

concepts = [row[:-1] for row in data]
target = [row[-1] for row in data]

def candidate_elimination(concepts, target):
    specific_h = concepts[0].copy()
    general_h = [['?' for i in range(len(specific_h))] for i in range(len(specific_h))]

    print("Initial Specific Hypothesis:")
    print(specific_h)

    print("\nInitial General Hypothesis:")
    for g in general_h:
        print(g)

    for i, h in enumerate(concepts):
        if target[i] == "Positive":
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = '?'
                    general_h[x][x] = '?'
        else:
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
                else:
                    general_h[x][x] = '?'
```

```
IDE Shell 3.14.3
File Edit Shell Debug Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

== RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 2 D.py ==
Initial Specific Hypothesis:
['Yes', 'Yes', 'Yes', 'Yes']

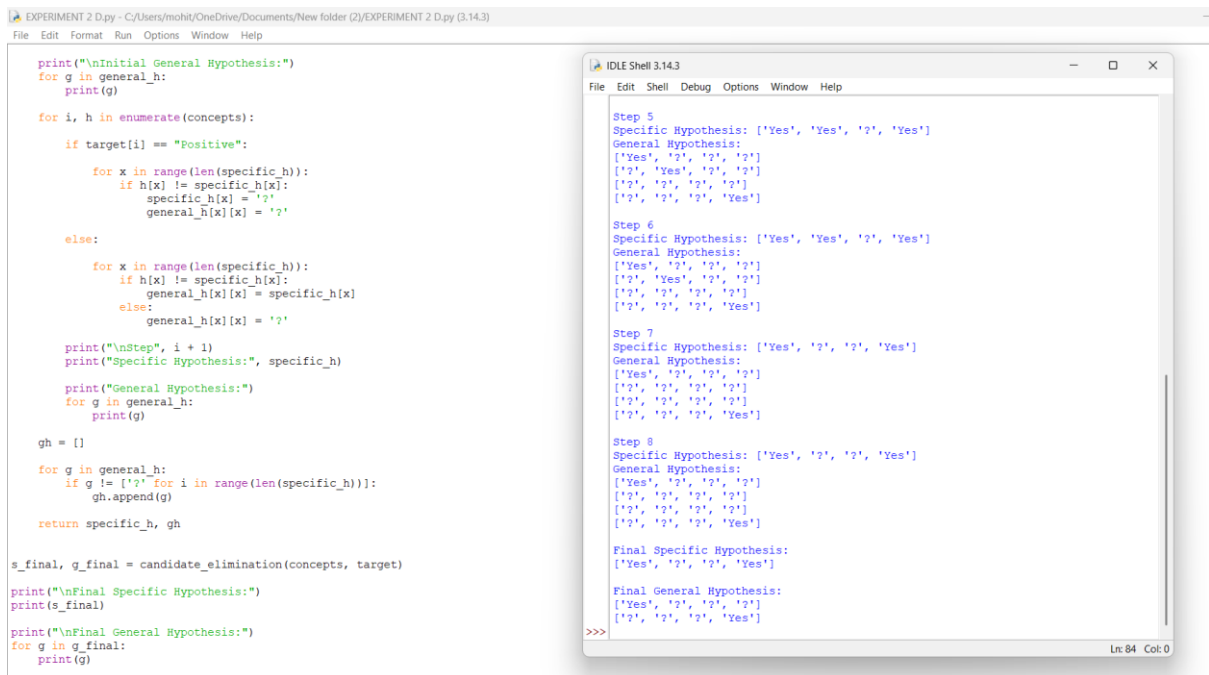
Initial General Hypothesis:
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']

Step 1
Specific Hypothesis: ['Yes', 'Yes', 'Yes', 'Yes']
General Hypothesis:
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']

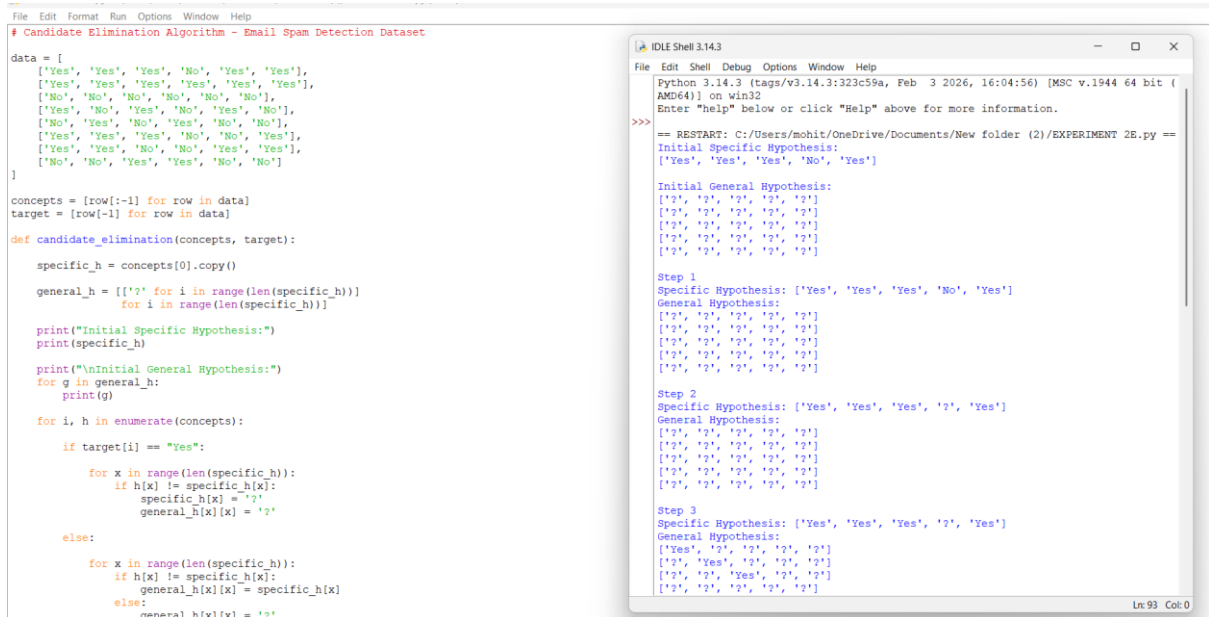
Step 2
Specific Hypothesis: ['Yes', 'Yes', '?', 'Yes']
General Hypothesis:
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']

Step 3
Specific Hypothesis: ['Yes', 'Yes', 'Yes', 'Yes']
General Hypothesis:
['Yes', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', '?']
['?', '?', '?', 'Yes']

Step 4
Specific Hypothesis: ['Yes', 'Yes', '?', 'Yes']
```



## DATASET 5





```

print("\nInitial General Hypothesis:")
for g in general_h:
    print(g)

for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                specific_h[x] = '?'
                general_h[x][x] = '?'
    else:
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'

print("\nStep", i + 1)
print("Specific Hypothesis:", specific_h)
print("General Hypothesis:")
for g in general_h:
    print(g)

gh = []

for g in general_h:
    if g != ['?'] for i in range(len(specific_h)):
        gh.append(g)

return specific_h, gh

s_final, g_final = candidate_elimination(concepts, target)

print("\nFinal Specific Hypothesis:")
print(s_final)

print("\nFinal General Hypothesis:")
for g in g_final:
    print(g)

```

```

['Yes', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', 'Yes']

Step 6
Specific Hypothesis: ['Yes', 'Yes', 'Yes', '?', '?']
General Hypothesis:
['Yes', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']

Step 7
Specific Hypothesis: ['Yes', 'Yes', '?', '?', '?']
General Hypothesis:
['Yes', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']

Step 8
Specific Hypothesis: ['Yes', 'Yes', '?', '?', '?']
General Hypothesis:
['Yes', '?', '?', '?', '?']
['?', 'Yes', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']
['?', '?', '?', '?', '?']

Final Specific Hypothesis:
['Yes', 'Yes', '?', '?', '?']

Final General Hypothesis:
['Yes', '?', '?', '?', '?']
['?', 'Yes', '?', '?', '?']

```

&gt;&gt;&gt;